

APPENDIX

SUSTAINABLE DEVELOPMENT GOALS (SDG) IMPACT

The CrimeLens platform is designed not only as a technical innovation but also as a contribution toward global sustainability and social stability. By leveraging Spatio-Temporal Graph Neural Networks and Real-Time Computer Vision, the project directly supports the following Sustainable Development Goals:

Goal 11: Sustainable Cities and Communities

- **Target 11.7:** Provide universal access to safe, inclusive, and accessible, green, and public spaces, in particular for women and children, older persons, and persons with disabilities.
- **Project Contribution:** CrimeLens enhances urban safety by providing law enforcement and community stakeholders with a normalized 0-100 Risk Index. By shifting from reactive policing to predictive, data-driven defense, the platform helps create more resilient and secure urban environments.

Goal 16: Peace, Justice, and Strong Institutions

- **Target 16.1:** Significantly reduce all forms of violence and related death rates everywhere.
- **Project Contribution:** The integration of a Real-Time Surveillance Module utilizing YOLOv11 for weapon detection and MobileNetV2 + LSTM for violence recognition allows for the immediate identification of lethal threats. This technology serves as a "digital guardian" to assist institutions in maintaining public order and reducing violent incidents.

Goal 9: Industry, Innovation, and Infrastructure

- **Target 9.5:** Enhance scientific research, upgrade the technological capabilities of industrial sectors in all countries.
- **Project Contribution:** CrimeLens bridges the "Integration Chasm" by synthesizing macroscopic predictive power with microscopic threat detection. The implementation of the Residual Spatio-Temporal Graph Attention Network (RST-GAT) represents a significant upgrade in the technological infrastructure available for public safety analytics.